
Second Order Impulse Response Verification

Given the transfer function

$$G(s) = \frac{(\zeta\omega_n)^2 + (\omega_d)^2}{(s + \zeta\omega_n)^2 + (\omega_d)^2} \quad (1)$$

the impulse response, $y_{\text{impulse}}(t)$, is the inverse Laplace transform of the transfer function, i.e.,

$$y_{\text{impulse}}(t) = \mathcal{L}^{-1}[G(s)].$$

We can write $G(s)$ in the form of the Laplace transform pair, using the *frequency shift theorem*,

$$\mathcal{L} \left[e^{-at} \sin(\omega t) \right] = \frac{\omega}{(s + a)^2 + \omega^2}.$$

We rewrite (1) as

$$G(s) = \left(\frac{(\zeta\omega_n)^2 + (\omega_d)^2}{\omega_d} \right) \frac{\omega_d}{(s + \zeta\omega_n)^2 + (\omega_d)^2}$$

and apply the inverse transform which yields

$$y_{\text{impulse}}(t) = \left(\frac{(\zeta\omega_n)^2 + (\omega_d)^2}{\omega_d} \right) \left(e^{-\zeta\omega_n t} \sin(\omega_d t) \right).$$

To simplify the constant, we use the relations

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

and

$$\omega_n^2 = (\zeta\omega_n)^2 + (\omega_d)^2,$$

which yields

$$y_{\text{impulse}}(t) = \left(\frac{(\zeta\omega_n)^2 + (\omega_d)^2}{\omega_d} \right) \left(e^{-\zeta\omega_n t} \sin(\omega_d t) \right) \quad (2)$$

$$= \left(\frac{\omega_n}{\sqrt{1 - \zeta^2}} \right) \left(e^{-\zeta\omega_n t} \sin(\omega_d t) \right) \quad (3)$$

$$= \left(\frac{\sqrt{(\zeta\omega_n)^2 + (\omega_d)^2}}{\sqrt{1 - \zeta^2}} \right) \left(e^{-\zeta\omega_n t} \sin(\omega_d t) \right). \quad (4)$$